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Unveiling the success of PIX: a Brazilian mobile instant payment system

Luiz Antonio Joia[1*], Manuela Fortes Lorenzo[1] and Tiago Azevedo Marot[1]

*Correspondence:
Luiz.joia@fgv.br

[1] Brazilian School of Public and Business Administration at Getulio Vargas Foundation, Rio de Janeiro, Brazil

Abstract

The proliferation of mobile technologies and the exponential growth of mobile subscriptions worldwide have led to significant opportunities for payment systems, particularly in low- and middle-income countries. Through online platforms, mobile instant payment systems have emerged as facilitators of financial inclusion and reduced banking transaction costs. In this vein, the success of a mobile instant payment system—PIX—developed by the Central Bank of Brazil presents an intriguing case for understanding the determinants of continued use intention of mobile payment apps. Therefore, data were collected from June 2023 to September 2023 through an online survey distributed via the authors' social networks and contacts, resulting in a final sample of 312 participants. Then, via structural equation modeling, it was set forth that perceived critical mass, trust, relative advantage, and affordable cost impact user satisfaction with PIX, which, in turn, influences users' intention to continue using PIX. This study thus presents empirical evidence and theoretical support to better understand the rapid and successful dissemination of PIX, providing valuable insights for policymakers, financial institutions, and researchers so that more features for this application can be designed and implemented, leading it to reach the entire potential Brazilian population and aiming to make it a benchmark for mobile instant payment systems in low- and middle-income countries.

Keywords: Mobile instant payment system, Mobile payment apps, PIX, P2P online payments, Continued intention to use information systems

Introduction

There are now more mobile subscriptions than people worldwide (WEF 2023). In fact, the proliferation of mobile technologies and the exponential growth of mobile subscriptions have created significant opportunities for payment systems (Bezovski 2016; Putrevu and Mertzanis 2024), mainly in low- and middle-income countries (LMICs) (Gachukia 2020). Furthermore, the financial sector is undergoing rapid change, driven by the emergence of fintechs (Bhatnagar 2022), which has created significant opportunities to increase financial inclusion in LMICs, such as Brazil, which faces high levels of financial exclusion (Joia and Cordeiro 2021). In this sense, the advent of new financial technologies can usher in a new era of sustainable development (Michael et al. 2022). Indeed, several scholars (Arner et al. 2020; Shankar 2024) argue that the advent of

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fintechs has provided policymakers in LMICs with tools that make it easier to achieve the United Nations Sustainable Development Goals (SDGs).

In addition, person-to-person (P2P) payments, facilitated through online platforms, have emerged as transformative solutions with the potential to reshape the financial landscape, reduce banking transaction costs and promote financial inclusion (Smolarsky 2021). However, despite increasing global financial inclusion (Ozili 2021), a substantial proportion of the world's adult population—approximately 1.7 billion individuals—still lack access to basic banking transactions. (World Bank 2022). Thus, addressing this issue is crucial, as financial services' affordability and accessibility play a pivotal role in poverty reduction and economic growth (Pazarbasioglu et al. 2020; Joia and Cordeiro 2021).

In line with that, the advent of a mobile instant payment app in Brazil—PIX—has garnered significant attention due to its rapid and widespread acceptance (BCB 2023a). Indeed, PIX, developed and implemented by the Central Bank of Brazil (BCB), has gained notable success in a relatively short span of time, offering innovative features and dramatically reducing financial transaction processing time (Julião, 2021). Its popularity among Brazilians has made it the most preferred payment method in the country (Zanobia 2023), particularly among microentrepreneurs (Negreiros 2023). Thus, the implementation of that system in the Brazilian context has attracted international attention due to its wide acceptance, achieved at an impressive speed (Sutto 2021).

Despite the remarkable success of PIX implementation, understanding its impressive diffusion trajectory and continued usage intention still needs further investigation. Therefore, finding the antecedents to the intention to continue using PIX is crucial to maintaining this sustainable success (Caldeira et al. 2023).

In line with that, this research intends to contribute to the extant scientific literature on mobile payment systems—which has predominantly focused on developed countries (Humbani and Wiese 2019)—by providing empirical evidence and theoretical support regarding the determinants of continued use of mobile payment apps in Brazil—an LMIC. In fact, in LMICs, electronic money and digital wallets have become the main financial solutions for the previously unbanked (Rosli et al. 2022). As an example, one can mention the use of mobile phones for financial transactions in some African countries, such as Kenya and Tanzania, or the deployment of mobile payment systems as part of a financial inclusion policy in Latin American countries, such as Peru (Reiss 2018).

However, despite successful and remarkable mobile payment and financial services cases, such as M-Pesa in Kenya (Kingiri and Fu 2020), Alipay and WeChatpay in China (Parasol 2022), GCash in the Philippines (Soquiat et al, 2026), and UPI in India[1] (Pal et al. 2019, 2020, 2021; Sharma 2019), global mobile payment adoption has lagged behind expectations, especially in developing regions, including Jordan (CliQ) (Khraim et al. (2011), Iran (Pol) (Hanafizadeh et al. 2014), and other emerging markets (Bailey et al. 2022; Qatawneh et al. 2025). The same phenomenon has been observed in some developed countries, such as Italy (TIPS) (Moroni et al. 2015) and the US (FedNow)

[1] In total, UPI payments grew by as much as 45% in 2023, accounting for 84% of all electronic payments in India. While this growth rate is impressive, the share of real-time payments in the total value of electronic transactions just slightly exceeds 10%, indicating there remains ample room for growth (Górka 2025).

(Carbó Valverde and Kahn 2016).[2] These facts demonstrate the relevance of understanding the impressive success in the implementation and dissemination of PIX in Brazil—the main objective of this research.

To achieve that objective, this work proposes a model that integrates established approaches related to the success of information systems (IS) implementation, aiming to identify the main factors that influence the intention of continued use of PIX. To address this, the study conducted an online survey with 312 participants and analyzed the data using structural equation modeling (SEM), aiming to answer the following research question: what factors drive the continued intention to use PIX in Brazil? The aim is to provide valuable insights for public policymakers, professionals from financial institutions and academics that can lead to increased financial inclusion and improved user experience in mobile payment systems.

It is important to stress that this study focuses on demand-side (user-centric) antecedents of the continuance intention to use PIX. While supply-side factors (e.g., system design, pricing models enabled by economies of scale) provide the essential infrastructure for mobile apps such as PIX (Kawamoto et al. 2023), in this work, they are considered boundary conditions for user behavior concerning the system under analysis.

This article is structured as follows. After this introduction, the theoretical foundation and the ensuing research hypotheses are presented. Next, the methodological procedure adopted is explained. After this, the results obtained are discussed in a subsequent section. Finally, the conclusions are presented, addressing the academic, managerial and public policy implications accrued from this work, its limitations and the next steps to be followed in that research area.

Literature review and research hyphoteses

Peer-to-peer mobile payment systems

The global adoption of mobile payment solutions is increasing due to their inherent convenience and several transactional advantages (Kaur et al. 2020). This proliferation is particularly significant in LMICs, where mobile payments are viewed as a catalyst for economic development and empowerment (GSMA 2016). In this context, P2P systems pose a substantial challenge to traditional payment methods—such as cards and cash—in everyday transactions (Belanche et al. 2022). Moreover, the landscape of P2P payments has undergone a transformative shift, significantly accelerated by the global COVID-19 pandemic (Abdullah and Khan Naved 2021). Recognizing this growing trend, financial institutions have been actively promoting collaborative ventures with fintechs (Palmié et al. 2020; Luo et al. 2022).

P2P payments can be summarized as money transfers between individuals (Markendahl and Apanasevic 2013). More specifically, P2P mobile payment is described as an "online technology that allows customers to transfer funds from their bank account or credit card to another individual's account via the internet or mobile phone" (Lara-Rubio et al. 2021, p. 2). In the past, P2P payments were limited to in-person transactions due to the need for physical exchanges of money (Kleivene 2018). However, the evolution of

[2] More comparative information between instant payment systems can be found at Górka (2025) and Haigh (2025), the latter comparing Brazil (PIX) and India (UPI).

online P2P payments has allowed customers to transact with other individuals through their online banking systems, intermediary platforms—such as PayPal—or mobile applications on their smartphones (Chandra et al. 2010; Kleivene 2018).

Although P2P mobile payment systems offer an efficient solution for casual transactions between two individuals—being prevalent in private transactions between acquaintances, friends or family—they are progressively invading the commercial domain (Irimia-Diéguez et al. 2023). Its adoption and continued use, however, have varied significantly from one country to another, with China leading this ranking (Taylor 2023), although barriers still need to be overcome via public policies, such as promoting digital financial product usage, designing risk-controlled digital financial products for customers who are excluded from traditional finance, and launching initiatives promoting financial literacy (Shen et al 2020).

Instant payment systems: the global landscape

Instant payment systems (IPSs) have transformed global financial infrastructure by enabling 24/7 real-time transfers, reducing cash dependency, and promoting financial inclusion (Giraldo-Mora 2023), mainly in emerging markets (Singh 2017; Shen et al. 2020; Falaiye et al. 2024). With over 70 countries implementing IPS by 2025, transaction volumes continue growing exponentially, fundamentally reshaping how money moves across economies (Górka 2025).

According to Sarkar and Rabha (2025), the Asia–Pacific region leads in IPS innovation, exemplified by India's unified payments interface (UPI), which has processed 89.5 billion transactions annually since its 2016 launch by the national payments corporation of India (NPCI). UPI's success stems from mandatory interoperability across banks and fintech apps such as PhonePe and Google Pay, combined with zero-cost transactions for users and merchants. The system supports advanced features, including recurring payments (UPI AutoPay) and offline transactions (UPI 123PAY). In contrast, China's system, operated by the people's bank of China (PBOC), processes transactions through QR codes and mobile apps such as Alipay and WeChat Pay but lacks UPI's unified national infrastructure (Chen 2025).

According to Lobo et al. (2025), Brazil's PIX system demonstrates rapid adoption potential in Latin America, achieving 84% adult penetration within three years of its 2020 launch by the BCB. The system processed BRL 118.4 billion[3] in a single day in 2024, driven by mandatory interoperability for large banks and competitive merchant fees ranging from 0.22 to 0.33%. PIX's centralized directory (DICT) ensures system uniformity while supporting payment aliases such as phone numbers and emails.

On the other hand, and according to Grabowski (2025), European systems face greater fragmentation, with SEPA instant credit transfer supporting euro-denominated transfers across 36 countries but struggling with voluntary participation, and the UK's Faster Payments Service, a 2008 pioneer in 24/7 transfers, remains constrained by legacy system compatibility requirements (Górka 2025).

[3] Approximately USD 22.1 billion as of December 1, 2025.

In North America, IPS adoption lags behind despite multiple initiatives. The Federal Reserve's FedNow and the Clearing House's RTP enable instant settlements but face voluntary adoption challenges, particularly among smaller institutions. Moreover, Zelle—processing 1.7 billion transactions in the first half of 2024—remains bank-centric with limited interoperability compared to global leaders (Górka 2025).

Finally, in Africa and the Middle East, Kenya's M-Pesa pioneered mobile money for financial inclusion but operates outside traditional banking networks (Abbas 2025), while government-backed systems such as Saudi Arabia's Sarie and UAE's IPI face infrastructure gaps in rural areas (Al-Asmari and Al-Arram 2025).

That said, below one delves deeper into PIX, the Brazilian instant mobile payment system, the focus of this research.

PIX: a P2P instant mobile payment system in brazil

In Brazil, the P2P instant mobile payment system—PIX—has achieved significant growth and penetration across various segments of the population within a short period of time (BCB 2023a). Indeed, the remarkable speed of a financial operation conducted by PIX makes it stand out, reducing financial transaction times from days to virtually seconds. This efficiency has made PIX the preferred payment method among Brazilians (Zanobia 2023) and a favored option for microentrepreneurs in the country (Negreiros 2023). In fact, the rate of adoption and use of PIX in Brazil was faster than that of other instant mobile payment systems implemented around the world (Sutto 2021), leading PIX implementation in Brazil to be a game changer for financial transactions in the country.

PIX was implemented within the country's payment system by the BCB on November 6, 2020 (Silva 2023). According to the BCB (2023b, p.1),[4] "PIX is the Brazilian instant payments system [...] where funds are transferred between accounts in a matter of seconds, at any time or day. It is practical, fast and safe. Payments using PIX can be made from a current account, savings account, or prepaid accountt." Thus, PIX stands out as one of the most significant innovations in the banking sector in recent decades Lorenzo et al. (2026). In its first month of operation, PIX registered 49.4 million users, representing 48% of Brazil's economically active population (Guimarães 2020). In October 2023, three years after its launch, this number almost tripled, reaching 161 million users signed up to the service (BCB 2023d). According to the BCB (2023d), in 2022, PIX was responsible for the inclusion of 71.5 million users in the financial system and was the most used payment method in the country in terms of the number of financial transactions.

PIX is a payment system that allows the transfer of funds in a few seconds, serving as an alternative to existing payment methods at the time of its launch.[5] Another feature is that PIX has no time or day-of-week limits, as it operates 24 h a day, 7 days a week, between any banks, from bank to fintech, from fintech to payment institution, and so on.

[4] Translated from Portuguese by the Authors.

[5] According to BCB (2023c), at the time of the introduction of PIX, Brazil had three types of credit transfers: TED (Express Wire Transfer), DOC (Credit Transfer Document), and TEC (Special Credit Transfer). According to Resolution BCB n°256, fund transfer via TED/TEC must be credited to the beneficiary within 90 min from the moment the debit is made to the sender's account. However, via DOC – which is no more available – the funds were only available for withdrawal on the following day (T + 1).

This makes PIX a fast and reliable payment option for transferring money. In addition to its speed, PIX is also a convenient payment method, as to transfer funds all the clients need is their phone numbers, emails, or CPFs (Brazilian Social Security Numbers). This eliminates the need for complete account details, making it easier and more convenient to send and receive payments. Additionally, with PIX, the necessity for financial intermediaries between accounts has been eradicated. Transactions can occur through three main methods: registered keys, known as PIX Keys, QR Codes, and, in the future, Near Field Communication (NFC) technology. According to the BCB (2023b), PIX offers several advantages, such as:

- Increased payment and transfer speed;
- Improved market competitiveness and efficiency;
- Reduced transaction costs, being 100% free for individuals and charging lower fees for companies than those associated with traditional payment methods managed by banks or credit cards;
- Increased security and improved customer experience;
- Financial inclusion for those who do not have a bank account, as it is mandatory to have one to use the system. Payments via PIX can, therefore, be made through digital accounts or payment platforms in Brazil (such as Mercado Pago or Pic Pay), thus requiring fewer documents and bureaucracy than via traditional banks.

As P2P instant mobile payments continue to gain prominence and fintechs take advantage of their vast potential, it is paramount to comprehensively explore the antecedents to the unprecedented success of this mobile app in Brazil, as presented below.

Approaches to successful IS implementation

Successful IS implementation has been extensively studied by academics and practitioners for a long time (Joia and Altieri 2018) and under different theoretical approaches (Rogers 2003; Fishbein and Ajzen 1975; Ajzen 1991; Davis et al. 1989). The technology acceptance model (TAM), proposed by Davis et al. (1989), along with its sundry variants, is one of the models most used to do that (Benbasat and Barki 2007). However, despite the importance of TAM and the several models derived from it, they all face several criticisms (e.g., Lee, Kozar, and Larsen, 2003; Benbasat and Barki 2007; Joia et al. 2016; Joia and Altieri 2018). This criticism arises mainly from the inability of these models to incorporate alternative theoretical frameworks for a more comprehensive understanding of technology adoption (Verkijika and Neneh 2021). Furthermore, these models have been criticized for not considering the unique characteristics of each technology; instead, they tend to treat all technologies as equivalent, thereby assuming that a single theoretical approach can explain their adoption (Orlikowski and Iacono 2001; Shih and Chen 2013; Lew et al. 2020). Critics (Lee, Kozar, and Larsen, 2003; Legris et al. 2003; Benbasat and Barki 2007, Belanche et al. 2022; Liébana-Cabanillas et al. 2020) argue that these limitations restrict the power of TAM and its variants to explain a greater percentage of the variance associated with the intention to use an IS (Legris et al. 2003; Joia et al. 2016; Joia and Altieri 2018). This issue suggests that essential factors may have been overlooked in these analyses (Liao et al. 2009).

Moreover, TAM and its variants focus on initial user adoption rather than the intention to continue using a system. Indeed, there has been a shift in the scientific literature on IS acceptance, with the focus shifting from measuring technology adoption to measuring its continued use (Ding et al. 2012; Shaw et al. 2018). As the objective of this work is to verify the antecedents of the continued intention to use the PIX application, one must look beyond the technology acceptance model (TAM), thereby exploring alternative models or frameworks that specifically address the user's postadoption behavior, which leads to continued intention to use the system (Saravanos et al. 2022). According to Shaw et al. (2018, p. 205), "…it appears that continued technology use is not just a continuation of technology adoption, but a phenomenon within itself. This raises additional questions regarding the suitability of TAM and successive extensions when measuring continued use." In fact, Shaw et al. (2018) developed an integrated model to predict continued IS use (TIM) by evaluating previous theoretical models. However, future research needs to empirically test the proposed model, which still lacks a standardized set of measures and indicators to be effectively operationalized by other researchers (Shaw et al. 2018).

In line with that and due to its potential to address these abovementioned issues, user satisfaction with the system can serve as a critical antecedent of IS continued use or even as a proxy for users' attitudes and intentions to continue using the system, as supported by Mather et al. (2002), Liao et al. (2009), Joia et al. (2016), Joia and Altieri (2018), Shaw et al. (2018), Caldeira et al. (2023), and Legramante et al. (2023). Therefore, one adopted user satisfaction with PIX as an antecedent for the continued intention to use this mobile app, in line with the recent works of Zhou (2013), Windasari et al. (2021) and Sharabati et al. (2022).

That said, the sundry variables associated with the continued intention to use an IS are consolidated below, with the aim of adequately explaining the success of PIX in Brazil.

Wired user style

User style refers to distinct patterns of technology interaction shaped by individuals' digital literacy, cognitive preferences, and behavioral tendencies. It is often dichotomized into wired (tech-savvy users with high digital integration) and nonwired (users with limited digital proficiency or resistance to technology adoption) (Liébana-Cabanillas et al. 2018). This classification emphasizes how users' intrinsic capabilities and attitudes influence their engagement with digital systems. Thus, wired users typically exhibit confidence in navigating digital interfaces, while nonwired users may experience anxiety or skepticism, leading to divergent adoption behaviors (Venkatesh et al. 2003).

In that vein, Araújo and Diniz (2021) found that mobile payment users with high digital literacy remain using the system after six months, attributing the same to their ability to leverage the system's features. Indeed, their intrinsic motivation and trust in technology align with Rogers' (2003) "early adopters," who drive diffusion through advocacy.

On the other hand, Caldeira et al. (2023) observed that users with low digital literacy (i.e., nonwired users) stopped using PIX in three months, on average, citing challenges such as security concerns and interface complexity. These barriers reflect technostress—a psychological state where technology demands exceed users' adaptive capacity (Liébana-Cabanillas et al. 2018). Thus, without interventions (e.g., simplified tutorials),

nonwired users' discontinuation can undermine PIX's inclusivity goals (BCB 2023b). Based on that, one can formulate H1, as presented below:

H1: Wired user style positively influences the continued intention to use PIX.

Perceived critical mass

Olson (1989) developed collective action theory (CAT), in which rational individuals continually evaluate the behaviors of their peers to guide their own choices toward cooperation (DeMarrais and Earle 2017). In this vein, the perceived critical mass (PCM) concept is derived from the CAT, proposing that a small number of early adopters of a new idea, technology, or innovation can influence others to join in, causing the movement to grow exponentially over time (Oliver et al. 1985). Indeed, as supported by Cheng et al., (2020), the likelihood of IS adoption and continued intention to use increases when users believe that most of their colleagues are using the system—in other words, a critical mass is perceived by them.

Perceived critical mass is also the "tipping point" to guarantee the self-sustainable future of a system (He et al. 2021). In fact, perceived critical mass is fundamental to ensuring the continued use of an IS (Li et al. 2018; Lynn et al. 2019), especially during its initial implementation phase—as in the case of P2P lending systems (Evans and Schmalensee 2010)—insofar as users are satisfied with the same. Thus, the lack of a critical mass of users is seen as a barrier to the adoption and continued intention to use mobile payment systems (Liébana-Cabanillas et al. 2016). Thus, perceived critical mass has a relevant effect on the intention to continue using an IS (Zhou et al. 2015; Lee and Kim 2020).

Zhou et al. (2015) conducted a study based on flow theory, which revealed a significant correlation between the size of a critical mass network and the level of perceived enjoyment and satisfaction in using an IS. Flow is described by Getzels and Csikszentmihalyi (2017) as a state of pleasurable experience through which an individual is completely focused and immersed in a task, to the point that nothing else seems to matter. This concept has become crucial in understanding user behavior related to IS adoption and continued intention to use. In fact, favorable user experience leads to higher levels of satisfaction and engagement, as observed by Calvo-Porral et al. (2017). Therefore, perceived critical mass can be considered an antecedent to user satisfaction with an IS, as supported by Lee and Kim (2020) and Zhou et al. (2015). Based on that, one can formulate H2, as presented below:

H2: Perceived critical mass positively influences user satisfaction with PIX.

Affordable costs

Numerous studies have highlighted the importance of affordable costs for user satisfaction with mobile payment applications (Humbani and Wiese, Humbani and Wiese 2019; Kim et al., 2010; Carlin et al. 2023; Boden et al. 2020; Qu et al. 2022; Kar 2021). Indeed, drawing from behavioral decision theory, the cost–benefit relationship is pivotal

in shaping individuals' behaviors and choices (Wu and Wang, 2005). According to Amoroso and Chen (2017), cost includes not only equipment costs but also access fees and transaction charges. Moreover, Mallat (2007) suggests that high costs associated with mobile payments discourage consumers from adopting and continuing using them, especially when there are no commensurate benefits or satisfaction. Other studies reinforce this idea by showing that consumers are less likely to keep using apps that are perceived as expensive and risky (Dini et al. 2018; Sarkar et al. 2020). Indeed, unless service providers are able to reduce transaction costs and increase the transparency of mobile payment transactions, consumers' perception of the high expenses associated with mobile payment systems will likely hamper or impede their adoption and continued intention to use (Zhou 2011). This perspective is shared by Chong (2013), who advises service providers not to price their services beyond competing offerings, especially considering that some competitors provide similar services for free. Moreover, Qu et al. (2022) argue that users will become satisfied with a system when the cost associated with its use is considered affordable by them.

In addition, the relationship between PIX's demand-side affordability and its supply-side economies of scale forms a self-reinforcing virtuous cycle, namely, mandatory zero-cost access for users triggers explosive adoption, which in turn generates an overwhelming scale that permanently lowers the system's operational costs per transaction, locking in its affordability, funding continuous innovation, and rendering any fee-based alternative fundamentally uncompetitive (Lobo and Brandt 2021).

Based on that, one can formulate the following hypothesis H3:

H3: Affordable cost positively influences user satisfaction with PIX.

Relative advantage

Relative advantage refers to the extent to which an innovation is perceived as offering greater benefits compared to its predecessor (More and Benbasat, Moore and Benbasat 1991). Several studies (Kalinic et al. 2019; Liu and Yi 2017; Shen et al. 2017; De Kerviler et al. 2016) have identified relative advantage as an important antecedent to the intention to adopt and continue using mobile payment systems. In this sense, the positive relationship between relative advantage and user satisfaction is explained by IS theories. DeLone and McLean's IS Success Model (DeLone and McLean 1992, 2003) establishes that relative advantage is a central element of system and information quality—constructs that directly determine user satisfaction. In parallel, task-technology fit theory (Goodhue and Thompson 1995) clarifies this mechanism: a system whose capabilities adapt to user tasks creates a relative advantage, which in turn increases perceived usefulness and ultimately leads to greater user satisfaction.

Compared to previous Brazilian payment instruments—DOC (settlement and fees on D + 1), TED (fast, but limited to business hours and often subject to fees), intrabank transfers (instant, but restricted to the same institution), *boletos* or bills of exchange (noninstant confirmation, D + 1/D + 2), and cards (merchant fees and deferred settlement, e.g., credit in T + 30)—Pix seems to have offered clear and immediate advantages, such as real-time settlement 24 h a day, 7 days a week, 365 days a year, generally zero

cost for individuals, lower acceptance costs for merchants, simplified addressing through "keys" (phone, email, CPF, or random key), interoperable QR codes between institutions, and broad ecosystem interoperability. These features may explain Pix's relative advantage over legacy options.

That way, one can formulate hypothesis H4:

H4: Relative advantage positively influences user satisfaction with PIX.

Complexity

Complexity is defined as the degree to which an innovation can be considered difficult to understand and use (Rogers 2003). In line with this, scholars have identified that complexity has a negative effect on user satisfaction regarding continued intention to use mobile payment systems (Al-Jabri and Sohail 2012; Joia and Altieri 2018).

The negative impact of complexity on user satisfaction can be explained by cognitive load theory (Sweller 1988), which posits that a complex system imposes a high external cognitive load—mental effort devoted to understanding the interface and navigation rather than the task itself. This unnecessary effort can lead to frustration and mental fatigue (Park and Kim, 2024). Furthermore, according to the expectation-confirmation theory (Bhattacherjee 2001), users expect an IS to be a tool that simplifies their work. A complex system that induces cognitive overload fails to confirm these expectations of ease of use, resulting in negative disconfirmation. Indeed, this negative disconfirmation is a primary antecedent to user dissatisfaction. Empirical studies have consistently supported this link, showing that complexity-induced cognitive load is a key driver of decreased satisfaction with IS (e.g., Lowry et al. 2014; Liu et al. 2025).

Based on that, hypothesis H5 is proposed:

H5: Complexity negatively influences user satisfaction with PIX.

Trust

Numerous definitions of trust exist, underscoring the intricate nature of this construct (Joia and Altieri 2018). The most prominently referenced definition of trust within the academic literature, as embraced in this study, stems from Mayer et al. (1995). According to them, trust entails two actors—one entrusting and one entrusted—signifying "the willingness of a party to be susceptible to the actions of another party based on the anticipation that the latter will undertake a specific action of significance to the trustor, irrespective of the ability to oversee or regulate that party" (Mayer et al. 1995, p. 712).

In fact, trust plays a pivotal role in fostering enduring relationships between consumers and providers (Liébana-Cabanillas et al. 2021). Furthermore, it stands out as one of the factors widely examined within the theory of intention to continue using IS, as indicated by various studies (Lai et al. Lai 2017; Kar 2021). Scholars have proposed that trust represents a user's willingness and propensity to remain loyal to a specific product or service, rooted in the anticipation of positive outcomes and performance associated with using that service (Joia and Altieri 2018). Indeed, a lack of user trust is the biggest

obstacle to the success of mobile financial systems in the long term (Gao and Waechter 2017). Thus, in the context of mobile payment systems, trust plays a fundamental role in mitigating uncertainty and strengthening users' satisfaction (Chin et al. 2022) since users' intention to continue using the system is influenced by the corresponding perception of risk and uncertainty associated with its use (Belanche et al. 2014). Thus, one can formulate H6:

H6: Trust positively influences user satisfaction with PIX.

User satisfaction

User satisfaction is defined as "the overall opinion and experience that a user feels" (Liébana-Cabanillas et al. 2021, p. 7), standing out as a predominant and extensively utilized metric for assessing IS success (Petter et al. 2013; Legramante et al. 2023). Furthermore, various authors have regarded user satisfaction as a factor influencing the continued intention to use mobile payment apps, as supported by Hsiao et al. (2016) and Nguyen et al. (2023). Indeed, numerous studies in the mobile payment services literature have established that satisfaction is paramount in shaping users' intentions to continue using those services (Zhou 2013; Humbani and Wiese 2019; Tang 2019; Yan et al. 2021).

Furthermore, successful IS implementation stems from users' intention to continue using it after initial adoption (Bhattacherjee 2001; Joia et al. 2016; Joia and Altieri 2018). Indeed, examining only the initial intention to use a mobile payment app or a P2P payment system is insufficient, as individuals may employ these technologies once and then discontinue their use (Tam et al. 2020). Thus, scholars argue that delving deeper into the continued intention to use an IS over time is crucial, as it is this intention that ultimately leads to its successful implementation (Revels et al. 2010; Lankton and McKnight 2012; Zhou 2013; Joia et al. 2016; Joia and Altieri 2018; Franque et al. 2023).

Thus, as the intention to continuously use PIX was chosen as the dependent variable of the developed model, the following hypothesis H7 is formulated:

H7: User satisfaction positively influences the continued intention to use PIX.

Based on what was presented, a model for the continued intention to use PIX is developed, as depicted in Fig. 1.

Method

Participants and procedures

Data were collected through an online survey distributed via the authors' social networks and contacts. The questionnaire was made available from June 2023 to September 2023. Attention checks were implemented to ensure that respondents were attentively responding to the survey. In total, 426 individuals participated in the research; however, only 359 respondents completed it to the end. Additionally, 47 individuals were disqualified based on the attention checks, resulting in a final sample of 312 participants.

A confirmatory factor analysis was performed to assess the validity of the measurement model. Thus, the calculation of the average variance extracted (AVE) and

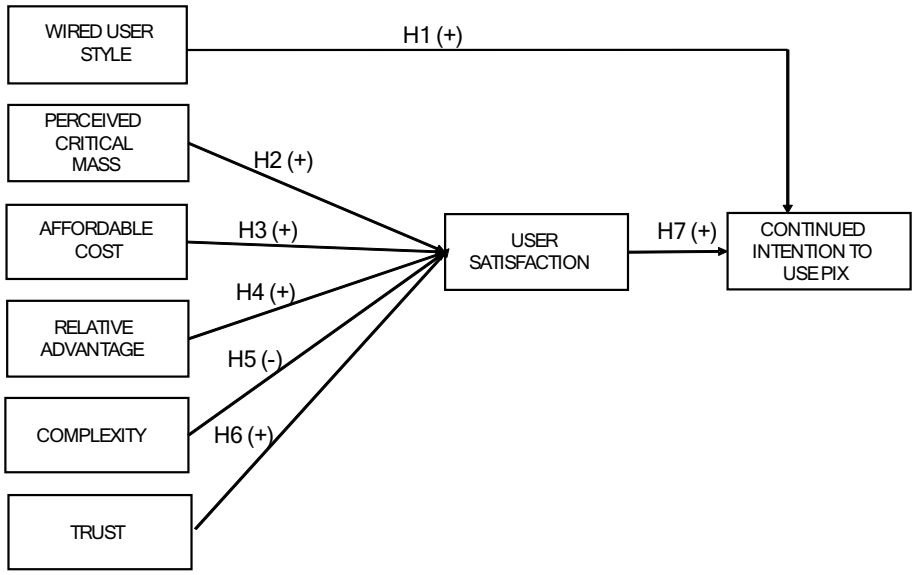


Fig. 1 Theoretical model

correlation analysis were conducted to test the convergent and discriminant validity, respectively. Moreover, composite reliability (CR) and Cronbach's alpha were computed to assess the reliability of the constructs. Finally, structural equation modeling was employed to verify the structural fit of the proposed model to the data through the following indices: chi-square (χ^2), standardized root mean squared residual (SRMR), root mean square error of approximation (RMSEA), Tucker–Lewis Index (TLI), and comparative fit Index (CFI). All analyses were performed in RStudio.

Measures

The survey included items adapted from previously validated scales found in the extant scientific literature. However, some items were modified to better fit the specific context of PIX. As the scales were originally validated in English and the survey was administered in Brazil, a three-step translation process was employed to translate the items into Portuguese. Initially, one of the authors performed the translation of the questions, followed by a meticulous review conducted by a second author. Subsequently, external researchers, who were unaware of the research objectives, conducted a final review of the translated items. All scales were measured via a five-point Likert-type scale, ranging from 1=totally disagree to 5=totally agree. All items of the abovementioned scales are made available in Appendix A. Table 1 shows the constructs and the references used for the scales.

Unlike what occurs with the variables quoted above, there is still some disagreement on how to measure user satisfaction with an information system. According to Ajzen and Fishbein (1980), Wixom and Todd (2005), Liu and Guo (2008), and Al-Jabri and Sohail (2012), user satisfaction can be understood as an attitude toward an object—in this case, an information system. Doll and Torkzadeh (1991) have thus developed a specific scale to measure user satisfaction with IS. However, Chin and Lee 2010) later challenged it, as they support the fact that satisfaction accrues from the difference between

Table 1 Constructs and references

Constructs	References
Wired user style	Crowley et al. (1992); Constantinides et al. (2004)
Perceived critical mass	Widyanto et al. (2022)
Affordable cost	Wu and Wang, (2005)
Relative advantage	Al-Jabri and Sohail (2012)
Complexity	Al-Jabri and Sohail (2012)
Trust	Al-Jabri and Sohail (2012)
User satisfaction	Bhattacherjee (2001); Hsiao et al. (2016); (Akdim et al. 2022)
Continued intention to use	(Akdim et al. 2022)

the system's actual performance and expected performance—the expectation-disconfirmation approach (Oliver 1980). However, Au et al. (2002) support that the expectation-disconfirmation approach, largely used in IS user satisfaction research, is not fully adequate to measure this construct. As there is to date no consensus on how to measure user satisfaction, it was decided to assess user satisfaction with PIX in this research via a Likert-type scale adapted from Bhattacherjee (2001) and Hsiao et al. (2016), as proposed by Akdim Akdim et al. (2022)

Results

Measurement model

The demographics of the sample (N = 312) are as follows: 36% of the sample is male**,** and the average age of the sample is 39.6 years (SD = 12.4). Moreover, 71% of the respondents have a university degree, with 37% of the sample holding a postgraduate degree. In addition, 47% of the respondents reported an income greater than BRL 7100 reais (approximately USD 1200). Moreover, 75% of the respondents have used PIX since its deployment in 2020. Finally, 57% of the respondents used PIX at least once a week, 24% every day and 19% every month. Table 2 depicts the descriptive statistics and the correlation matrix between variables.

To assess common method variance, Harman's single-factor test was used. The results indicated no significant bias, as more than one factor had eigenvalues greater than 1, and the first factor accounted for only 29% of the variance (Podsakoff et al. 2003).

Structural model

A structural equation modeling (SEM) analysis was conducted to investigate the predictive power of the model on user satisfaction with PIX and continued intention to use the same. The results yielded adequate fit indices: χ^2 (383) = 522.94 ($p < 0.001$), CFI = 0.986, TLI = 0.984, SRMR = 0.061, RMSEA = 0.034. Fig. 2 illustrates the coefficients of the structural model.

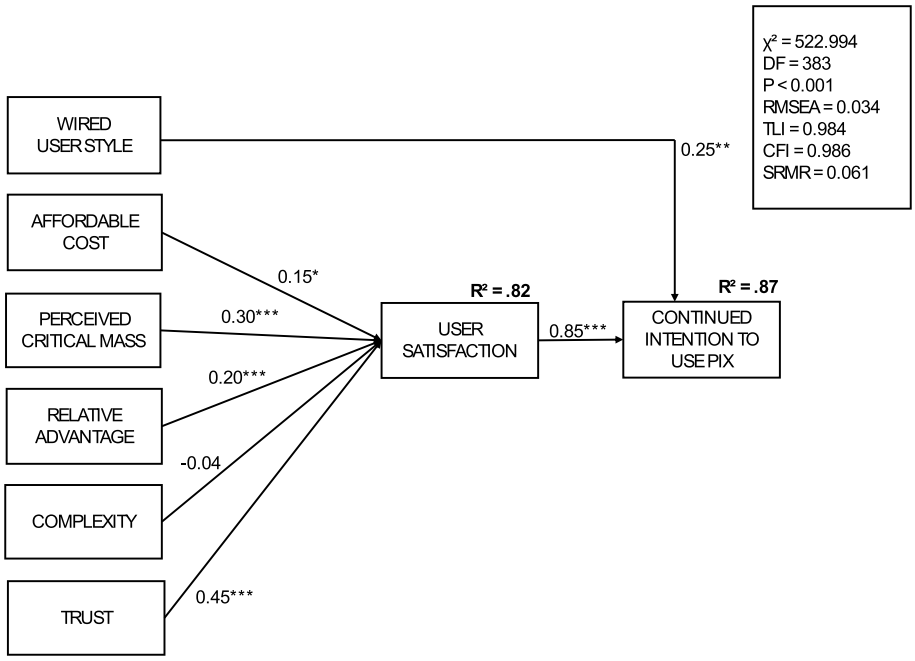
Table 2 Descriptive statistics and correlation matrix

	Scale	M	SD	1	2	3	4	5	6	7	8	CR	α
Correlation table among investigated variables													
1. Wired user style	1–5	4.32	0.06	**0.33**								0.66	0.47
2. Affordable cost	1–5	4.33	0.69	0.05	**0.57**							0.78	0.54
3. Perceived critical mass	1–5	4.73	0.47	0.19***	0.34***	**0.74**						0.92	0.83
4. Relative advantage	1–5	4.32	0.76	0.05	0.31***	0.39***	**0.69**					0.9	0.82
5. Trust	1–5	4.47	0.62	0.03	0.32***	0.41***	0.52***	**0.69**				0.92	0.86
6. User satisfaction	1–5	4.8	0.47	0.09	0.39***	0.52***	0.48***	0.59***	**0.85**			0.96	0.88
7. Continued intention to use	1–5	4.8	0.45	0.17**	0.34***	0.55***	0.44***	0.48***	0.72***	**0.91**		0.97	0.89
8. Complexity	1–5	1.87	0.93	0.01	− 0.02	− 0.08	− 0.18**	− 0.14*	− 0.11	− 0.13*	**0.64**	0.84	0.76

N = 312

The values in bold print placed in the diagonal correspond to the average variance extracted (AVE)

CR = Composite Reliability; α = Cronbach's Alpha



(***) p <1%; (**) p<5%; (*) p<10%

Fig. 2 SEM results. (***) $p < 1\%$; (**) $p < 5\%$; (*) $p < 10\%$

Table 3 Test of hypotheses

Hypothesis (Hi)	Outcome	*p* value
H1: Wired user style influences positively the continued intention of using PIX	Supported	*p* value < 5%
H2: Perceived critical mass positively influences user satisfaction with PIX	Supported	*p* value < 1%
H3: Affordable cost positively influences user satisfaction with PIX	Supported	*p* value < 10%
H4: Relative advantage positively influences user satisfaction with PIX	Supported	*p* value < 1%
H5: Complexity negatively influences user satisfaction with PIX	Non-supported	–
H6: Trust positively influences user satisfaction with PIX	Supported	*p* value < 1%
H7: User satisfaction positively influences the continued intention to use PIX	Supported	*p* value < 1%

The results demonstrate that perceived critical mass, relative advantage, and trust have a significant and positive effect on user satisfaction with PIX ($p < 0.010$), followed by wired user style ($p < 0.05$) and, to a lesser extent ($p < 0.10$), affordable cost. In addition, complexity has no significant (negative) impact on user satisfaction with PIX. Furthermore, user satisfaction showed a positive and significant relationship with the continued intention to use the application ($p < 0.01$).

These outcomes are discussed in the next section.

Discussion

Based on the outcomes accrued from the hypothesis tests, Table 3 is presented to consolidate the results obtained, which are discussed throughout this section.

The results obtained herein reveal that trust plays a significant role as an antecedent to users' satisfaction with PIX. In fact, in recent decades, numerous studies have focused

their attention on the influence of trust on user satisfaction with new payment systems (including mobile payment systems) and their subsequent adoption and continuous use (Franque et al. 2023). For example, n Xin et al. (2013) emphasize that trust substantially impacts consumers' inclination to use continuously mobile payment systems. Similarly, Duane et al. (2014) propose that trust is the most influential factor in people's intention to continuously use their smartphones for payments. Furthermore, Cao et al. (2016) argue that among all the factors that influence users' intention to adopt and keep using a mobile payment system, trust stands out as the most significant. Last, Chin et al. (2022) highlighted that trust in mobile payments has a remarkably robust influence on consumers' satisfaction and willingness to use it continuously. That way, the results duly tally with the extant scientific literature on the subject.

It is important to stress that PIX was developed by the BCB and was implemented on a large scale in Brazil in 2021 after a period of tests was conducted throughout 2020. The mobile app has been supported, improved and regulated by the BCB, which gives it a high level of trust among citizens (Libba et al. 2021). It is also important to highlight that the large-scale implementation of PIX took place during the period of social isolation of the population caused by the COVID-19 pandemic, which led the BCB to reinforce the security of the system to strengthen the population's trust in it (do Vale et al. 2022), considering its relevance at the time it was released.

Another significant antecedent to users' satisfaction with PIX is perceived critical mass. As already mentioned, perceived critical mass is defined as the point on the adoption curve of an innovation at which it is self-sustaining (Rogers 2003); namely, later adopters of an innovation are influenced by earlier adopters of the same. However, when examining the deployment of a new IS, Markus (1987) realized that its early adopters can also be influenced by its later adopters, as well as by those who choose not to adopt the same. For example, the first users of PIX faced challenges and limitations, as few people used the application at that time, which prevented the first users from carrying out financial transactions very often. In fact, in the early stages of PIX implementation, there was uncertainty about who else was using PIX and who would adopt it in a subsequent stage, as was the case with other new technologies when they were made available—for example, when the email was adopted by organizations (Markus 1994). As a result, awareness of how many other people are using PIX impacts individuals' intention to use and keep using it, as the app was designed primarily to carry out financial transactions between two persons (P2P). This result also complies with other research on that subject, such as the work of Byrd et al. (2021)on the role of perceived critical mass on the continued intention to use mobile healthcare apps.

Relative advantage is another significant antecedent of users' satisfaction with PIX and their consequent willingness to continue using it. According to Rogers (2003), relative advantage is the extent to which an innovation is perceived as better than the preceding idea. The findings on the positive impact of relative advantage on users' satisfaction with PIX and their consequent intention to continue using the same are consistent with other studies related to the continued intention to use mobile payment systems (Arvidsson 2014; Yunus 2015; Mensah 2021). In fact, these results suggest that in relation to traditional ways of carrying out financial transactions, the relative advantage of PIX is based on the possibility of payments regardless of time and location, remote and almost

ubiquitous access to payment services, and the end of the need to queue at banks and make payments in cash, as was also observed by Mallat (2007) and Kuriakose et al. (2022) regarding mobile payments in general.

Additionally, users' wired styles have a significant positive impact on their willingness to continue using PIX. Indeed, empirical studies on mobile payments corroborate that digitally immersed users (namely, wired users) adopt innovations faster and sustain usage when systems align with their tech-savvy behaviors (Dahlberg et al. 2015; Oliveira et al. 2016). In Brazil, PIX's success among wired users—a growing demographic in the country's digitally connected population—has been pivotal to its rapid adoption (BCB 2020). These users validate PIX's reliability, creating a feedback loop where continued use reinforces its indispensability. Thus, being "wired" (digitally connected) predicts continued PIX usage because it reflects a user base primed for digital solutions. Their fluency, performance demands, and ingrained digital habits make PIX a natural extension of their daily routines, while their social influence accelerates broader acceptance. Thus, a wired user style can be considered a significant antecedent of continued intention to use PIX.

The last and less statistically significant factor related to users' satisfaction with PIX and the consequent continued intention to use the same was affordable cost. At first sight, this result seems to be at odds with reality. However, there is a simple explanation for that. Although the use of PIX is free, banks—after the implementation of PIX by the BCB—also stopped charging their customers for financial transactions carried out via traditional channels. As the sample adopted is biased toward middle- and upper-class respondents with access to other free payment methods, that context can explain the low significance of affordable cost in user satisfaction with PIX and the consequent intention to continue using it. Further research can investigate whether those results change for low-income individuals.

However, there is a factor that did not have a significant impact on user satisfaction with PIX and their subsequent intention to continue using it, namely, complexity. Regarding this variable, although PIX seems complex at first glance, its continuous use makes it simple to understand, which impacts the lesser relevance of this factor in user satisfaction with the app, which is supported by the learning curve throughout time (Liébana-Cabanillas and Lara-Rubio 2017). Indeed, Hackbarth et al. (2003) and Carter and Belanger (2005) argue that the complexity scale captures the same effects of the perceived ease of use scale of TAM. In line with that, Legris et al. (2003, p. 195) listed a series of works that did not find perceived ease of use to be a significant antecedent to IS adoption and continued use (see, for example, Hu et al. 1999 and Joia and Altieri 2018). Davis et al. (1989) also support that the learning curve associated with an IS lessens the importance of this construct over time. Likewise, Venkatesh et al. (2003) and Kavak and Anwar (2019) argue that perceived ease of use tends to be more relevant during the first stages of learning how to use an IS, becoming less important throughout the period of use. In other words, studies on the continued use of a system suggest that user familiarity and effective system performance outweigh initial complexity concerns (Bhattacherjee 2001; Limayem et al. 2007). In addition, training and iterative system improvements further reduce complexity's role in sustained IS usage (Hong et al. 2006). In the sample, all respondents had already used PIX, and most had a university degree and belonged

to the upper income bracket in Brazil. Therefore, they likely had good digital literacy and reasonable experience with the system, which may have reduced the importance of complexity as an antecedent to satisfaction with PIX (Venkatesh et al. 2003; Nurdin et al. 2019; Luthfia et al. 2021).

Conclusions

This study sheds light on the main factors that influence the intention to continue using the instant payment system PIX. The findings highlight that perceived critical mass, trust, and relative advantage are the main factors contributing to user satisfaction with the system ($p < 0.01$), which in turn is a strong antecedent of continued intention to use PIX ($p < 0.01$). Furthermore, the paper highlights that users' style (wired) is a significant antecedent ($p < 0.05$) for their intention to continue using PIX, which indicates that the success of such initiatives also depends on users' digital literacy, highlighting the importance of digital inclusion for the success of such endeavors. Moreover, the cost affordability of PIX in an LMIC such as Brazil has a moderate impact ($p < 0.10$) on user satisfaction with the app, which, as stated, is a strong antecedent of continued intention to use the system. Thus, the free nature of PIX has been a significant antecedent of its rapid and widespread acceptance as well as continued use among citizens, mainly in lower income brackets (Lobo and Brandt 2021).

Several studies on PIX adoption in Brazil can be found using different theoretical approaches—see, for example, Amboage et al. (2024) (applying UTAUT) and Felipe et al. (2023) (applying UTAUT2). In addition, some studies have investigated the antecedents of user satisfaction with PIX, although they do not relate the same to the continued intention to use the app (e.g., Valcanover et al. 2022). There are also studies that address the important role of the BCB in the success of PIX in the country, such as Lobo and Brandt (2021). However, to date, few studies have addressed the antecedents of continued PIX use, which demonstrates the relevance of this work. Indeed, Caldeira et al. (2023) developed a study that investigated the antecedents of continued PIX use in Brazil using a user satisfaction-based model, as in this study. They also found a strong causality between user satisfaction with PIX and the intention to continue using it, as in this study. However, they did not consider antecedents considered important in this research, such as perceived critical mass, "wired" user style, relative advantage, cost affordability, and trust. Instead, they chose to base their model primarily on TAM variables, namely, perceived ease of use and perceived usefulness, which ended up leading to R^{2s} lower than those found in this study.

There are also some studies comparing PIX with other mobile payment systems implemented worldwide. Aurazo and Gasmi (2024) compared four mobile instant payment systems implemented in different countries, namely, PIX (Brazil), UPI (India), M-Pesa (Kenya), and Yape (Peru). For Aurazo and Gasmi (2024), PIX and UPI are considered the most successful initiatives in the world in this arena, which indicates the relevance of comparing them. Indeed, by any measure, PIX is perceived as an enormous success in Brazil (Schapiro et al. 2023), as is UPI in India– also considered a remarkable initiative in this country (Sarkar and Rabha 2025), in addition to being considered by some scholars a benchmark in the mobile payment systems landscape (Sheba and Vasanthi 2024). Interestingly, the success of both PIX in Brazil and UPI in India appears to lie in their

core design principles, which have led to the critical antecedents for their continued use (Pranger 2024), namely:

i) Strong Regulatory or Public Mandate: The top-down, mandatory approach of PIX (BCB) and the strong public–private push behind UPI forced widespread interoperability, preventing fragmentation and ensuring that the network effect could take hold immediately (Haigh 2025), leading users to a perceived critical mass and trust in the systems. This contrasts with the voluntary participation models of FedNow (US) and SEPA Instant (EU), which lead to slower, more fragmented adoption (Górka 2025);

ii) Radical User-Centric & Low-Cost Design: PIX and UPI were explicitly designed to be simpler and cheaper than existing options (cash, cards, etc.). Features such as aliases and QR codes have reduced the complexity of PIX, while zero-to-low fee structures have eliminated barriers to low-value person-to-person transactions, making these features essential for financial inclusion (Haigh 2025) as they provide perceived relative advantage and accessibility. Additionally, the importance of the "wired" user style for the success of both ventures can also be noted since financial inclusion essentially depends on digital inclusion (Vik et al. 2024).

Furthermore, the BCB's role as a nonprofit, mandated orchestrator was crucial to PIX's success. It created a level playing field where the main competition focuses on the quality of service and innovation offered by PIX, rather than ownership of the payment network itself (Schapiro et al. 2023). In this way, the scale achieved made the system socially efficient and financially sustainable, permanently consolidating its affordability advantage (Lobo and Brandt 2021).

That said, the main contributions of this study to both theory, practice and public policymaking are set forth below.

Contributions to academia, practice, and public policymaking

This study provides contributions to academia, financial institutions, and policymakers committed to refining the user experience with mobile payment systems and, as a consequence, strengthening financial inclusion in Brazil and similar emerging markets.

From an academic perspective, this research aims to be relevant by identifying factors related to satisfaction and continued use of PIX by combining established models of adoption and continued use of mobile payment systems. Indeed, the fusion of these theories with practice aims to enrich the understanding of the continued use of mobile payment systems, paving the way for future studies in this area. Furthermore, this work investigates how PIX brilliantly connects consumers, merchants, traditional banks, and fintechs (da Cunha Duarte 2025). Thus, this study provides a model to architect a platform to rapidly reach critical mass, analyzing the specific strategies that have enabled direct and indirect network effects. Furthermore, this work addresses the role of BCB as a catalyst for this initiative (Lobo and Brandt 2021), given that traditional innovation theory focuses on private companies (Öberg 2019). In this sense, PIX is an example of a powerful theoretical model of state-led digital disruption, providing insights into how a regulatory body (BCB) can successfully design, launch, and govern a public service

in the digital age—which is relevant for theorizing about the successful deployment of national mobile payment systems. Finally, this study provides empirical evidence on how a well-designed digital infrastructure can overcome known banking barriers to the inclusion of low-income clients (Kumar 2024), offering a replicable model for other emerging economies.

For financial institutions, this research can help them respond, leverage, and compete in a financial market transformed by disruptive public infrastructure. Indeed, with instant and free payments becoming a commodity, the study highlights the importance of fintechs in building a strategic ecosystem for PIX, as well as the importance of traditional banks shifting their strategies and business models to be part of this ecosystem (Sampaio and Ornelas 2024), by developing new value propositions for customers (e.g., linking PIX to investment or credit accounts). Finally, the work highlights the need for financial institutions to learn about the monetization strategies that have emerged around PIX (Padia 2025), such as facilitating payments for micromerschants, providing API solutions for SMEs, and developing advanced security and fraud prevention services.

For policymakers, the study provides a replicable policy framework for using public digital infrastructure to boost competition and economic modernization. Indeed, this research demonstrates how acting as a platform architect (creating public infrastructure)—rather than just a regulator—can enable central banks such as the BCB to stimulate massive competition and private-sector innovation in financial products and services (Ragazzo and Caminha 2025). In addition, PIX possesses a powerful architecture to drive unprecedented financial inclusion in Brazil by being low-cost, accessible, and efficient. Its greatest success has been in formalizing transactions and bringing a massive number of people and small businesses into the digital payment ecosystem (Lorenzo et al. 2025). However, its potential is capped by barriers that exist outside the financial system itself, namely, digital infrastructure, education, and cybersecurity (Lobo et al. 2025).

In 2024, it was estimated that 20.9 million in Brazil did not have a mobile phone for personal use, representing a small fraction of the low-income population in the country—this percentage was 18.6% in 2019 and 12.4% in 2023 (IBGE 2025). In fact, most low-income individuals do have access to at least a basic mobile device—either personally or through shared household or community use—which allows them to use PIX despite limited resources. What PIX enables, therefore, is not universal digital inclusion but relative financial inclusion by eliminating traditional barriers such as banking fees, minimum balance requirements, and the need for physical branches. Nonetheless, individuals without any phone access remain excluded, and this limitation highlights that the inclusiveness of PIX is contingent upon broader structural conditions of digital access. In other words, **while PIX has broadened financial inclusion for most low-income Brazilians, ongoing disparities in digital access prevent truly universal coverage.**

Therefore, for PIX to become a truly inclusive tool, its success must be measured not only by transaction volume but also by its reach to the most marginalized populations. This requires a concerted effort involving the following:

- BCB: supporting ongoing public education campaigns, promoting digital financial product usage, and strengthening security protocols.
- Government: investing in digital infrastructure (internet access) in underserved areas.
- Financial institutions: developing even simpler user interfaces and robust fraud prevention systems.
- Civil society: providing digital and financial literacy training at the community level.

Only by addressing these external barriers through public policies can PIX's full and positive impact on financial inclusion be achieved.

Research limitations and future steps

Some limitations present in this study should be addressed. First, the questionnaires were disseminated through the researchers' own social media networks. This explains why the sample is made up of individuals with a higher level of education (71% of the sample has higher education) and higher income, which does not represent ordinary citizens in Brazil. The sample, thus, is biased toward high-income and high-education users, which limits the generalizability of the results to the Brazilian population in general (but makes them useful for this segment of the country's population).Therefore, future research can extend this study to other samples, focusing particularly on marginalized segments of society due to the financial inclusion potential accrued from this P2P mobile payment system (Sampaio and Ornelas 2024).

Second, it is possible to notice the asymmetry of responses in the correlation table, i.e., all scales had average responses close to the maximum response point, and the standard deviation was systematically small. This pattern presents limitations that future research can address by refining measurement indicators and including individuals from a broader range of sociodemographic backgrounds.

Furthermore, despite the low reliability of Cronbach's α for the wired user style ($\alpha = 0.47$) and affordable cost constructs ($\alpha = 0.54$), it was decided to keep them in the model since the composite reliability (CR)—considered a better index to assess reliability (Peterson and Kim 2013)—of the latter is CR = 0.78 (considered acceptable) and of the former is CR = 0.66—tolerated in exploratory studies or early-stage research (Kline 2016). However, for future investigations, it is suggested to use more comprehensive scales for "wired user style" and "affordable cost".

In line with that, the "wired user style" construct presented an AVE of 0.33 (below 0.5) (Haji-Othman and Yusuff 2022). This points to the need to refine the respective scale to increase the convergent validity of this variable (Haji-Othman and Yusuff 2022). One solution is to replace the previous construct with the "user digital readiness" construct, as defined and measured by Nasution et al. (2018). According to those authors, "user digital readiness" means the inclination and willingness to migrate and adopt digital technologies and the readiness to create new innovative opportunities using this technology to achieve goals faster and with better results (adapted from, Nasution et al. 2018, p. 97).

In addition, future studies could develop experimental research to investigate to what extent the variables proposed in the model cause changes in satisfaction with PIX or in

the continued intention to use it. This would be of great importance, considering the lack of research on the causal antecedents of satisfaction with and continued use of PIX.

Another suggestion for future work is related to the measurement of real user behavior. Indeed, the inclusion of measures that covered actual user behavior would provide stronger external validity to the study. In other words, instead of using a subjective perception prone to various prejudices, future research could evaluate the user's own behavior, which would overcome the potential problems of self-reported measures, as well as provide an estimate closer to reality regarding the impact of antecedents in the target behavior.

Furthermore, this study focused on demand-side antecedents of PIX continuance intention of use. While factors such as perceived usefulness and trust are critical from the user's perspective, the ecosystem's success also relies on fundamental supply-side determinants. As noted in prior literature (e.g., Kawamoto et al. 2023), achieving economies of scale is a pivotal factor that enables low-cost infrastructure and sustainable business models for digital payment platforms. Future research could adopt a dual-perspective framework to empirically examine how such supply-side enablers interact with user perceptions to jointly influence the long-term sustainability and evolution of instant payment systems such as PIX.

Last, future research could take into account factors related to the characteristics and idiosyncrasies of users concerning the use of mobile applications in general—in addition to the wired user style considered in this study—as proposed by Joia and Oliveira (2010). For example, user risk aversion may have a relevant impact on the use of mobile payment applications (Long et al. 2023). In that future research, it would be possible to separate the external aspects of the mobile application—those related to the user's idiosyncrasies and traits—from the internal aspects of the app—those related to the app's design interface, ease of use, etc. In this way, relationships could be found between the different factors considered to understand their contributions to the completion of an effective financial transaction via PIX.

In sum, the transformative potential of mobile payment solutions in LMICs becomes clear when analyzing the success of PIX in Brazil, which can also be seen in other mobile payment apps, such as Bizum in Spain (Visonti-Caparrós and Campos-Blázquez, 2022; Liébana-Cabanillas et al. 2021) and M-pesa in Africa (Ndung'u, 2018; Rouse et al. 2023).

Financial inclusion is a key objective of the United Nations Sustainable Development Goals, which has led the BCB to prioritize the promotion of financial inclusion since 2010, incorporating it into its strategic objectives (BCB 2010). In this vein, the BCB has been implementing a comprehensive set of initiatives to democratize the country's financial services. In this regard, PIX has revolutionized payment methods in the country and paved the way for a more inclusive and digitally connected future (Sampaio and Ornelas 2024), notwithstanding its dependence on how "wired" the citizens are.

Appendix A

Survey items and related references

Instructions

Please indicate your level of agreement with the following statements based on your experience using PIX. Use the following scale:

1—Strongly Disagree, 2—Disagree, 3—Neutral, 4—Agree, 5—Strongly Agree.

User style (US)

Bellman et al. (1999); Joia and de Oliveira (2010); Chen and Demirci (2019); Choi and Park (2020).

1. **US1:** I frequently buy products on the internet.
2. **US2:** I frequently search for information on the internet.
3. **US3:** I spend more than 10 h a day on the internet.
4. **US4:** I use instant messaging apps (like WhatsApp

Relative advantage (RA)

Al-Jabri and Sohail (2012).

5. **RA1:** PIX is a more convenient way to make payments.
6. **RA2:** PIX allows me to make payments more efficiently.
7. **RA3:** PIX gives me more control over my payments.
8. **RA4:** Compared to other methods, PIX is useful for making my payments.

Affordable cost (COS)

Liu et al. (2019).

9. **COS1:** I use PIX because the transactions are free.
10. **COS2:** I use PIX because it uses very little of my mobile data.
11. **COS3:** I use PIX because my bank does not charge me to use it.

Complexity (CPX)

Al-Jabri and Sohail (2012).

12. 12. **CPX1:** Using PIX requires much mental effort.
13. 13. **CPX2:** Using PIX requires technical skills.
14. 14. **CPX3:** Using PIX can be frustrating.

Trust (TRU)

Liébana-Cabanillas et al. (2021).

15. **TRU1:** I believe PIX will keep its promises and commitments.

16. **TRU2:** PIX is trustworthy.
17. **TRU3:** I would describe the PIX payment system as honest.
18. **TRU4:** I think that mobile payment systems such as PIX act with integrity.

Perceived critical mass (PCM)

Widyanto et al. (2022).

19. **PCM1:** The people I have financial interactions with (buying or selling) use PIX.
20. **PCM2:** Many people I know want to continue using PIX in the future.
21. **PCM3:** Many of my colleagues make payments using PIX.
22. **PCM4:** I think a large number of my friends use PIX.
23. **PCM5:** People around me use PIX to make payments.

User satisfaction (SAT)

Akdim et al. (2022).

24. **SAT1:** I believe I made the right choice by using PIX.
25. **SAT2:** I am satisfied with using PIX.
26. **SAT3:** I am content with using PIX.
27. **SAT4:** Using PIX pleases me.

Continued intention to use (CIU)

Akdim et al. (2022).

28. **CIU1:** I intend to continue using PIX in the future.
29. **CIU2:** I expect to continue using PIX in the future.
30. **CIU3:** I will keep using PIX as regularly as I do now.

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Author contributions
L.A.J.—conceptualization, empirical design, theoretical background, interpretation of results, writing; M.F.L.—data collection, data analysis and interpretation of results, theoretical background, draft manuscript preparation, editing, and writing. T.A.M.—data collection, data analysis and interpretation of results. All the authors have read and approved the final manuscript.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
The study was approved by the Ethical Compliance Committee on Research Involving Human Beings—CEPH/FGV (Reference number: P.175.2023). All participants signed an informed consent form and were informed that they could withdraw from the study at any point.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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